

Tableau: Lecture Notes

1. Tableau Ecosystem

Tableau is an end-to-end data analytics platform. Its core components are designed for different stages of the analytics workflow.

- **Tableau Desktop/Public/Creator:**
 - **Desktop (Paid):** The main authoring tool for creating visualizations, dashboards, and stories.
 - **Public (Free):** A free version to create visualizations but **must** save workbooks to the public Tableau Cloud platform. Good for learning.
 - **Creator:** A licensing role that includes Tableau Desktop and Prep.
 - **Tableau Server/Cloud (Online):**
 - **Server (On-Premise/Private Cloud):** Used to publish, share, and manage workbooks/data sources within an organization.
 - **Cloud (SaaS):** The fully hosted version of Server. Allows users to view, interact with, edit, and collaborate on published content from a web browser or mobile app.
 - **Tableau Prep:** A separate application used for cleaning, shaping, and combining data before analysis in Desktop.
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2. Getting Started & Core Interface

The standard workspace is where you build your visualizations.

Area	Purpose
Data Pane	Lists all available fields from your data source, automatically categorized as Dimensions or Measures.
Columns/Rows Shelf	Fields dragged here create the columns and rows of your view (e.g., placing Sales on Rows and Year on Columns).
Marks Card	Controls the look and feel of the data marks (Color, Size, Label, Detail, Tooltip,

	Shape).
Filters Shelf	Used to narrow down the data displayed in the view.
Pages Shelf	Used to separate a view into a sequence of pages based on the values in a dimension.
"Show Me"	A panel that suggests appropriate chart types based on the fields currently selected in the Data Pane.

- **Dimensions** (Categorical Data): Typically fields that categorize or describe data, such as **Name, Date, Region**. They are often discrete (blue pills).
 - **Measures** (Quantitative Data): Fields that can be measured, aggregated, or calculated, such as **Sales, Profit, Quantity**. They are often continuous (green pills).
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3. Data Connections

You can connect Tableau to almost any data source.

- **Live Connection:** Tableau queries the data source directly.
 - **Pros:** Real-time data, saves space on your machine.
 - **Cons:** Performance depends on the source database/network speed; can slow down the source system.
 - **Data Extract (.hyper):** A compressed snapshot of the data, stored locally (or on the Server/Cloud).
 - **Pros:** Extremely fast performance, portable, supports offline work.
 - **Cons:** Data is not real-time (must be refreshed), requires local storage space.
 - *Best Practice:* Use extracts for large datasets or slow connections.
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4. Metadata Management

Metadata is "data about the data." It helps you control how Tableau interprets your raw source data.

- **Renaming Fields:** Give columns business-friendly names (e.g., change Cust_ID to

Customer ID).

- **Changing Data Type:** Ensure columns are correctly identified (e.g., **String, Number, Date, Boolean, Geographic Role**).
 - **Creating Aliases:** Rename specific *values* within a dimension (e.g., change CA to California).
 - **Creating Hierarchies:** Organize related dimensional fields for drill-down analysis (e.g., combine Region > State > City).
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5. Combining Data: Joins vs. Data Blending

These are two methods to bring data from multiple tables/sources together.

A. Joins (Recommended for Same Source)

- Combines tables from the **same data source** (or from compatible different sources using a **cross-database join**).
- Merges data **at the row level** (creates a single, merged table).
- Defined in the **Data Source tab** *before* creating a sheet.

Type	Description
Inner	Includes only rows that have matching values in both tables.
Left	Includes all rows from the left table, and the matching rows from the right table. Non-matches get null for right table fields.
Right	Includes all rows from the right table, and the matching rows from the left table.
Full Outer	Includes all rows from both tables. Non-matches get null where applicable.

B. Data Blending (For Different, Dissimilar Sources)

- Combines data from **separate, different Tableau data sources** (e.g., Excel and a SQL database).
 - Data is **not merged** at the row level; instead, Tableau queries each source **independently**, aggregates the results, and then visually presents them together in the view.
 - Requires a **Primary** data source (used first) and a **Secondary** data source.
 - A **linking field** (indicated by a chain icon ) is required to define the relationship.
 - *Limitation:* All fields from the secondary source are treated as aggregated values (using ATTR or another aggregation).
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6. Essential General Concepts

- **Worksheet:** A single page where you create an individual visualization (a viz).
- **Dashboard:** A collection of several related worksheets, filters, and other objects presented together to provide a holistic view.
- **Story:** A sequence of worksheets or dashboards that work together to convey a guided narrative about the data.
- **Calculated Fields:** Custom fields you create using a formula to extend your data (e.g., [Sales] - [Cost] = [Profit]).
- **Parameters:** Dynamic values that can replace constant values in calculations, filters, and reference lines. They allow the end-user to interactively control parts of the visualization.